
LM358 Op Amp: Datasheet, Circuit, Applications [Video]

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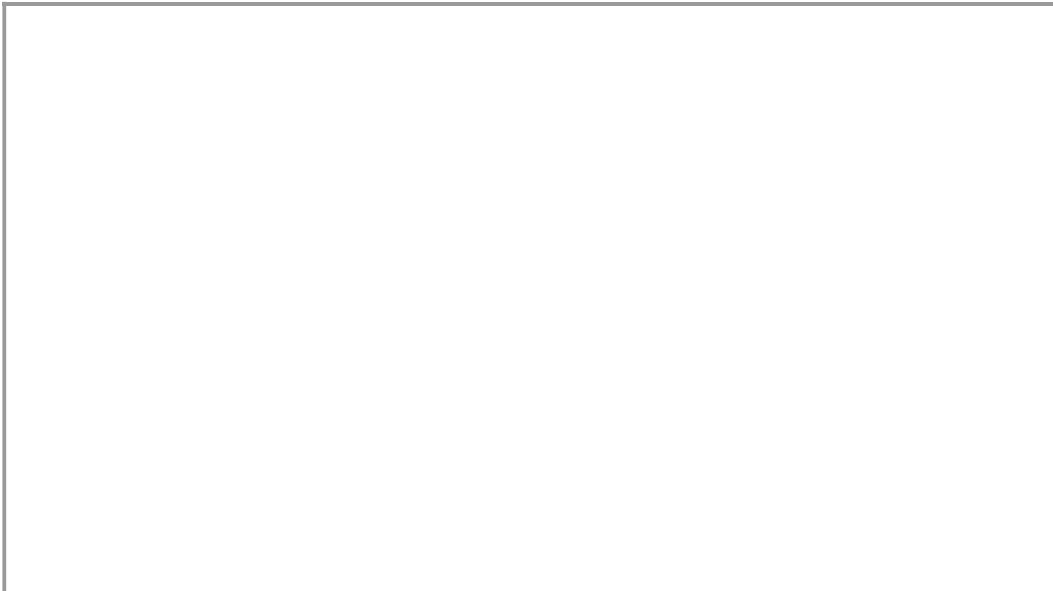
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Description

The **LM358** consists of two independent, high gain, internally frequency compensated operational amplifiers which were designed specifically to operate from single power supply over a wide range of voltages. Operation from split power supplies is also possible and the low power supply current drain is independent of the magnitude of the power supply voltage.

Application areas include transducer amplifiers, dc gain blocks and all the conventional op amp circuits.



TOP 5 Electronics Projects using LM358

Catalog

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LM358 CAD Model



V

LM358 Symbol



LM358 Footprint

LM358 Features

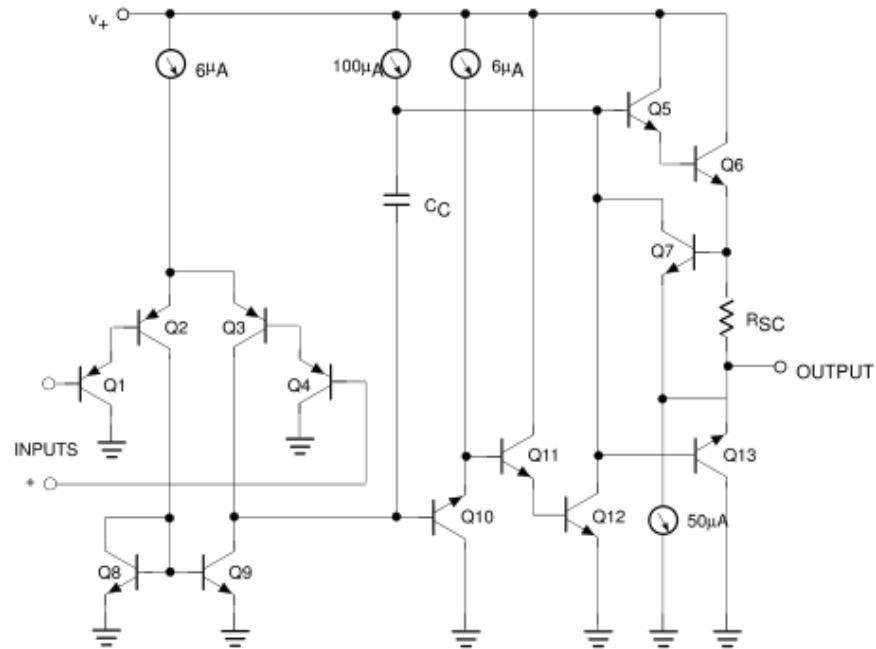
- Internally frequency compensation
- High DC voltage gain (about 100dB)
- Unit gain bandwidth (about 1MHz)
- Wide supply range of 3 V to 30V
- Dual power supply ($\pm 5 \sim \pm 15V$)
- slew-rate-limited (0.3V/us)
- Low power consumption, suitable for battery power supply
- Low input offset voltage and current
- Common-mode input voltage range includes ground, enabling direct sensing near ground
- Differential input voltage range is wide, equal to power supply voltage range
- Large output voltage range (0 to $v_{cc}-1.5v$)

LM358 Applications

- Merchant network and server power supply units
- Multi-function printers
- Power supplies and mobile chargers
- Motor control: AC induction, brushed DC, brushless DC, high-voltage, low-voltage, permanent magnet, and stepper motor
- Desktop PC and motherboard
- Indoor and outdoor air conditioners
- Washers, dryers, and refrigerators

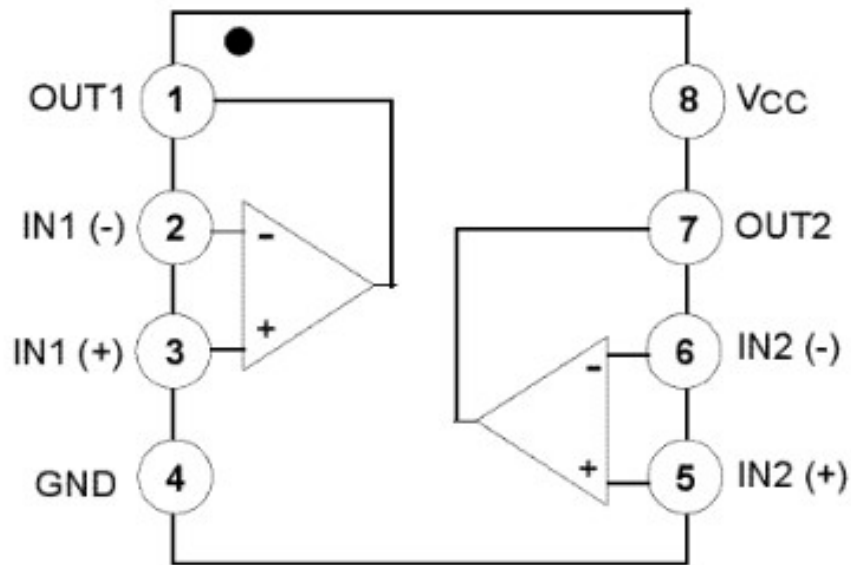
- AC inverters, string inverters, central inverters, and voltage frequency drives
- Uninterruptible power supplies
- Programmable logic controllers
- Electronic point-of-sale systems

LM358 Circuit



LM358 Opamp Electret Amplifier Circuit

LM358 Pinout



LM358 pin function: Pin 1 is the output, pin 2 is the inverting input, and pin 3 is the in-phase input; 4-pin is negative power supply (when dual power supply operates) or ground (when single power supply operates); Pin 5 is the in-phase input; Pin 6 is the inverting input; Pin 7 is the output; pin 8 is positive power supply; Pins 1, 2, and 3 are one op amp channel, and pins 5, 6, and 7 are another op amp channel.

LM358 Working Principle

Pin 8 is main power supply input. The voltage of pin 2 is compared with that of pin 3, and the voltage of pin 6 is compared with that of pin 5, corresponding to two independent outputs: 1OUT and 2OUT.

When 1IN+ is greater than 1IN-, 2IN+ is greater than 2IN-, 1OUT 2OUT outputs high level.

When 1IN+ is less than 1IN-, 2IN+ is less than 2IN-, 1OUT 2OUT outputs low level.

No pull-up resistor is required at the output of LM358. The output voltage range is: 0V~VCC-1.5V.

LM358 Parameters

Number of Channels (#)	2
Total Supply Voltage (Min) (+5V=5, +/-5V=10)	3
Total Supply Voltage (Max) (+5V=5, +/-5V=10)	32
GBW (Typ) (MHz)	0.7
Slew Rate (Typ) (V/us)	0.3
Rail-to-Rail	In to V-
Vos (Offset Voltage @ 25C) (Max) (mV)	7
Iq per channel (Typ) (mA)	0.35
Vn at 1kHz (Typ) (nV/rtHz)	40

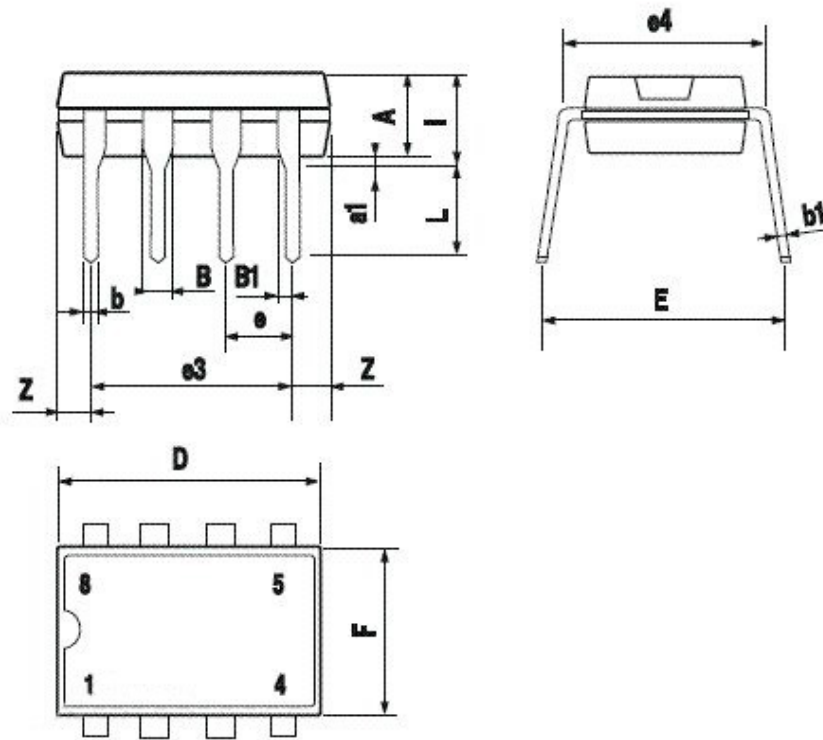
Rating	Catalog
Operating Temperature Range (C)	0 to 70
Package Group	PDIP SO SOIC TSSOP VSSOP
Offset Drift (Typ) (uV/C)	7
Input Bias Current (Max) (pA)	150000
CMRR (Typ) (dB)	80
Output Current (Typ) (mA)	30
Architecture	Bipolar

LM358 Electrical Characteristics

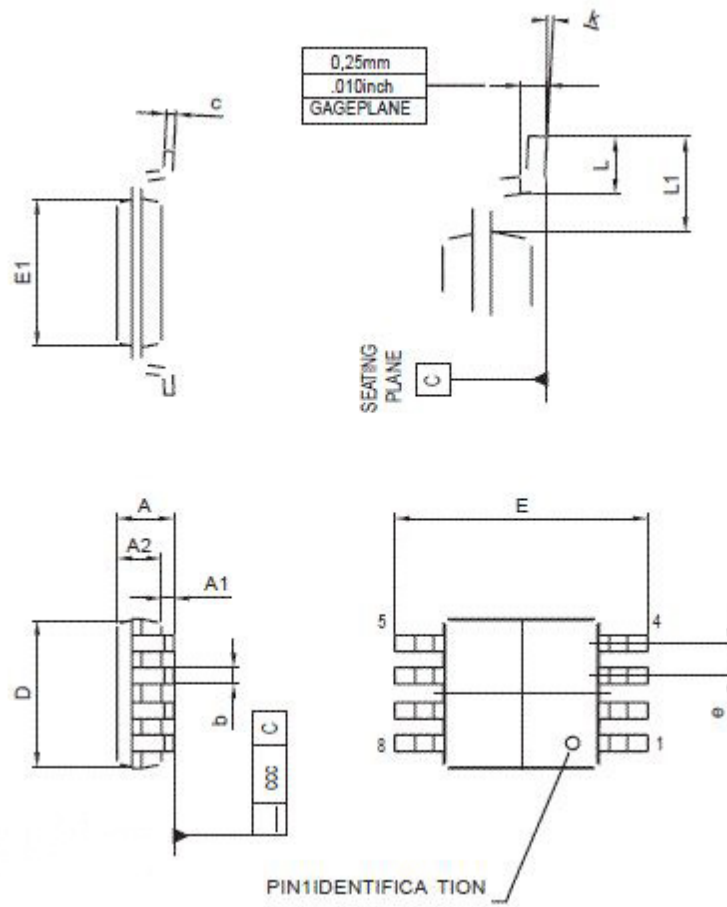
$V_{CC}=5.0V$, $V_{EE}=GND$, $T_a=25^{\circ}C$, unless otherwise specified

PARAMETER	SYMBOL	TEST CONDITIONS	MIN	TYP	MAX	UNIT
Input Offset Voltage	$V_{I(OFF)}$	$V_{CM}=0V$ to $V_{CC}-1.5V$ $V_{O(P)}=1.4V$, $R_S=0\Omega$		2.9	7.0	mV
Input Common Mode Voltage	$V_{I(CM)}$	$V_{CC}=30V$	0		$V_{CC}-1.5$	V
Differential Input Voltage	$V_{I(DIFF)}$				V_{CC}	V
Output Voltage Swing	V_{OH}	$V_{CC}=30V$, $R_L=2K\Omega$	26			V
		$V_{CC}=30V$, $R_L=10K\Omega$	27	28		V
	V_{OL}	$V_{CC}=5V$, $R_L \geq 10K\Omega$		5	20	mV
Large Signal Voltage Gain	G_V	$V_{CC}=15V$, $R_L \geq 2K\Omega$ $V_{O(P)}=1V \sim 11V$	25	100		V/mV
Power Supply Current	I_{CC}	$R_L=\infty$, $V_{CC}=30V$		0.8	2.0	mA
		$R_L=\infty$, Full Temperature Range		0.5	1.2	mA
Input Offset Current	$I_{I(OFF)}$			5	50	nA
Input Bias Current	$I_{I(BIAS)}$			45	250	nA
Short Circuit Current to Ground	I_{SC}			40	60	mA
Output Current	I_{SOURCE}	$V_{I(+)}=1V$, $V_{I(-)}=0V$ $V_{CC}=15V$, $V_{O(P)}=2V$	10	30		mA
	I_{SINK}	$V_{I(+)}=0V$, $V_{I(-)}=1V$ $V_{CC}=15V$, $V_{O(P)}=2V$	10	15		mA
		$V_{I(+)}=0V$, $V_{I(-)}=1V$ $V_{CC}=15V$, $V_{O(P)}=200mV$	12	100		mA
Common Mode Rejection Ratio	CMRR		65	80		dB
Power Supply Rejection Ratio	PSRR		65	100		dB
Channel Separation	CS	$f=1KHZ \sim 20KHZ$		120		dB

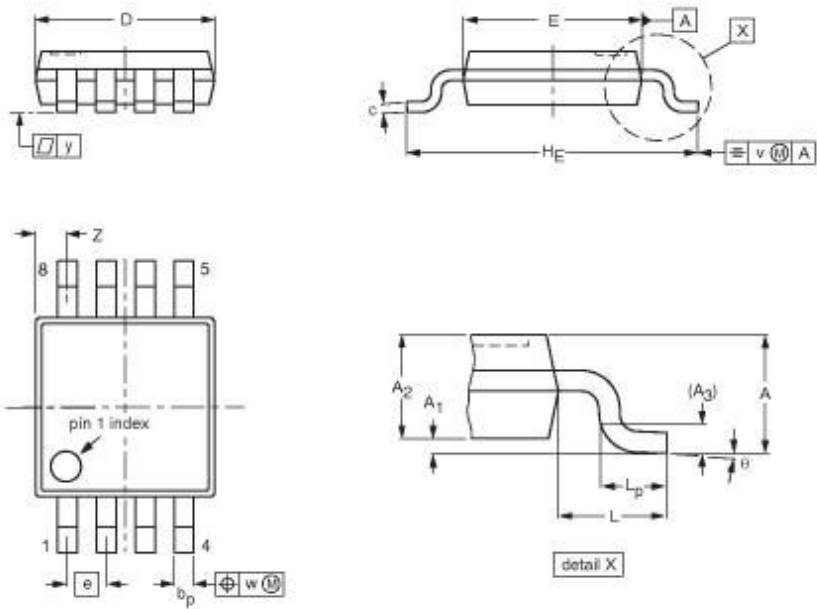
LM358 Package



LM358 DIP8 Package Diagram



LM358 SOP8 Package Diagram



DIMENSIONS (mm are the original dimensions)

UNIT	A max.	A ₁	A ₂	A ₃	b _p	c	D ⁽¹⁾	E ⁽²⁾	e	H _E	L	L _p	v	w	y	Z ⁽¹⁾	θ
mm	1.10	0.15 0.05	0.95 0.80	0.25	0.45 0.25	0.26 0.15	3.10 2.90	3.10 2.90	0.65	5.10 4.70	0.94	0.70 0.40	0.1	0.1	0.1	0.70 0.35	6° 0°

Notes

1. Plastic or metal protrusions of 0.15 mm maximum per side are not included.
2. Plastic or metal protrusions of 0.25 mm maximum per side are not included.

OUTLINE VERSION	REFERENCES				EUROPEAN PROJECTION	ISSUE DATE
	IEC	JEDEC	EIAJ			
SOT505-1						99-04-09

LM358 TSSOP8 Package

LM358 Product Compliance

USHTS	8542390001
ECCN	EAR99

LM358 Manufacturer

Texas Instruments Inc. (TI) is an American technology company that designs and manufactures semiconductors and various integrated circuits, which it sells to electronics designers and manufacturers globally. Its headquarters are in Dallas, Texas, United States. TI is one of the top ten semiconductor companies worldwide, based on sales volume. Texas Instruments's focus is on developing analog chips and embedded processors, which accounts for more than 80% of their revenue. TI also produces TI digital light processing (DLP) technology and education technology products including calculators, microcontrollers and multi-core processors. To date, TI has more than 43,000 patents worldwide.

LM358 Instruction

The output voltage of LM358 can only be $0 \sim 3.7\text{ V}$ not $0 \sim 5\text{ V}$ when IN^+ is input from $0 \sim 5\text{ V}$. In other words, when IN^+ input $0 \sim 3.7\text{ V}$, the voltage can follow OUT, when the input is greater than 3.7 V , the output will still be 3.7 V .

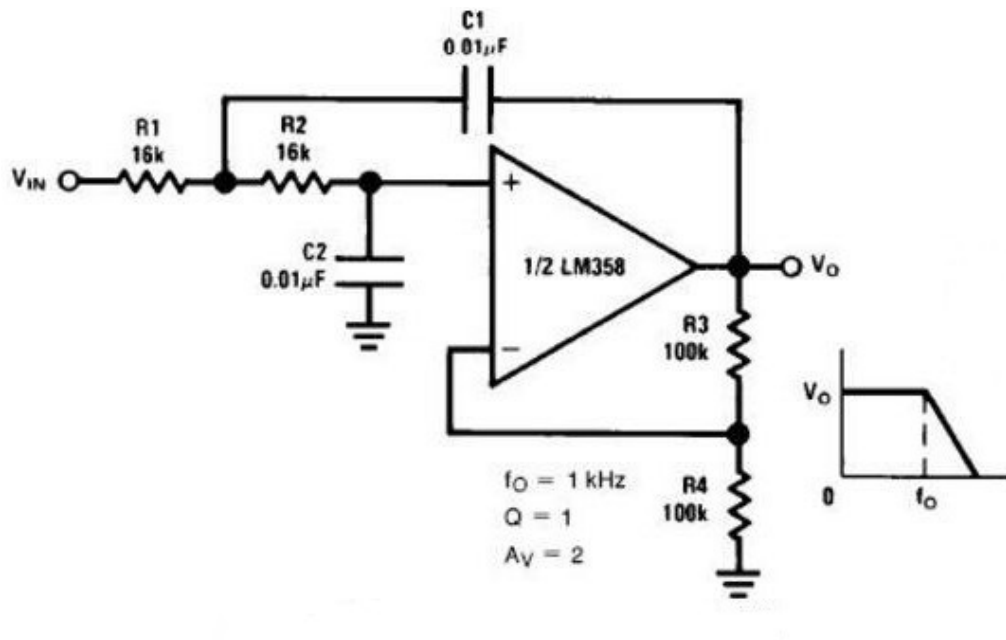
Because LM358 is an emitter output, the output range is at most $\text{VEE}+0.7 \sim \text{VCC}-0.7$. Like TL082, LM358 has no rail to rail performance, and the highest output is $3.X\text{ V}$.

Solution:

- (1) Change IC. instead of LM358, use a full amplitude voltage op amp, known as the rail to rail OP AMP in the industry. For example, the commonly used TLC2262 in industry has the same pin function as LM358, which means that the two can be interchanged.
- (2) To ensure that the output is within the working range of LM358, the full-scale output voltage can be designed 2 V lower than the power supply.

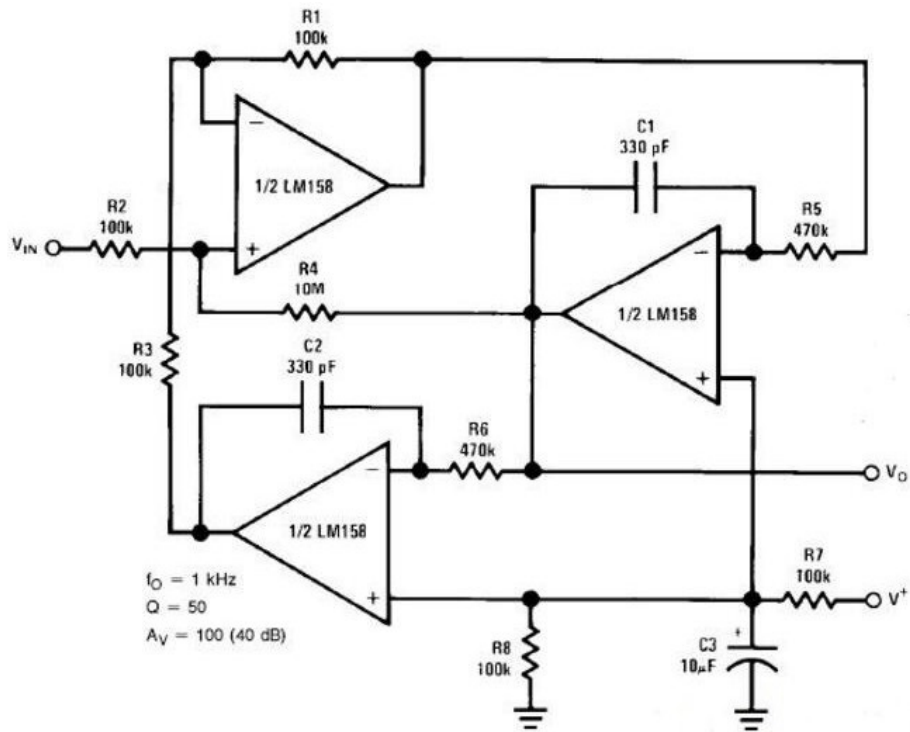
LM358 Application Circuit

DC Coupling Low-pass RC Active Power Filter



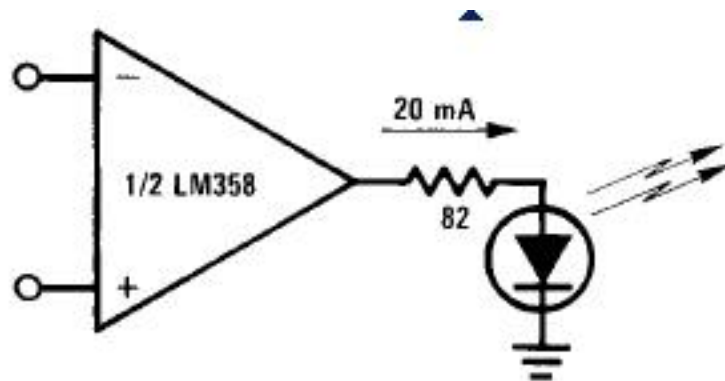
DC Coupling Low-pass RC Active Power Filter

RC Active Band-pass Filter



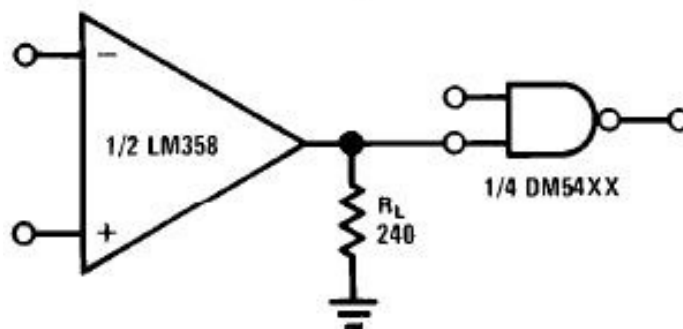
RC Active Band-pass Filter

LED Driver



LED Driver

TTL Drive Circuit



TTL Drive Circuit

Component Datasheet

LM358 Datasheet

FAQ

- **What is lm358 op amp?**

LM358 is a dual op-amp IC integrated with two op-amps powered by a common power supply. It can be considered as one half of LM324 Quad op-amp which contains four op-amps with common power supply. The differential input voltage range can be equal to that of power supply voltage.

- **What is lm358 used for?**

LM358 can be used as transducer amplifier, DC gain block etc. It has large dc voltage gain of 100dB. This IC can be operated on wide range of power supply from 3V to 32V for single power supply or from $\pm 1.5V$ to $\pm 16V$ for dual power supply and it also support large output voltage swing.

- **Why lm358 is used in IR sensor?**

IC Lm358 is used as a comparator when IR receiver senses IR radiations. When the o/p of lm358 goes high, then LED connected at the o/p turns ON. The output pin of the IC LM358 is used to interface with PIC microcontroller.

- **How does an lm358 work?**

IC LM358– LM358 consists of two independent, high gain operational amplifiers in one package. Important feature of this IC is that we do not require independent power supply for working of each comparator for wide range of power supply. LM358 can be used as transducer amplifier, DC gain block etc.

- **What is the difference between lm358n and lm358p?**

The suffix denotes the manufacturer's packaging code. The 'N' is used by most manufacturers for the plastic 8-pin package. The 'P' is used by a few manufacturers for the plastic 8-pin package. Raven Luni is right, there is NO difference between the two devices, just the mfgs.

- **How do I know if my lm358 op amp is broken?**

Measure the DC voltage at the +input. then measure the DC voltage at the output. if the results are significantly different, the opamp is most likely shot. if they are the same, the opamp is most likely ok and the problem is something else.

- **What is the difference between lm386 and lm358?**

The LM386 is a complete audio power amplifier, the LM358 is a dual operational amplifier. When using the LM358 e.g. as a pre-amplifier, you will have to supply a separate power amplifier.

• What is a dual op amp?

With a dual supply op amp, the $V+$ terminal of the op receives a positive voltage and the $V-$ terminal connects to negative voltage. Therefore, any input signal fed into the op amp can swing from the positive voltage supply to the negative voltage supply.

• What is LM series?

The following is a list of LM-series integrated circuits. ... The LM series originated with integrated circuits made by National Semiconductor. The prefix LM stands for linear monolithic, referring to the analog components integrated onto a single piece of silicon.

• How to import lm358 into LTspice?

1. Download model file and unzip.
2. Place .cir file in same folder as schematic.
3. Place "opamp2" symbol on schematic.
4. Change "opamp2" value to LMX58_LM2904.
5. Place directive on schematic ". lib LMX58_LM2904. CIR" without quotes.



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